

July 7, 2026

RE: Senior Quant Researcher - CTA/Short-Term, Squarepoint

Dear Hiring Team,

I am writing to apply for the Senior Quant Researcher - CTA/Short-Term role at Squarepoint. My background combines Python-based quantitative research projects, large-scale time-series data systems, production monitoring, and the engineering discipline needed to move ideas from analysis into reliable automated workflows. I am particularly interested in Squarepoint's focus on researching intraday trading opportunities, questioning statistical robustness, and comparing live strategy behavior with simulations.

At RBC, I architected an institution-wide telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash, ingesting 150 TB/day across 10,000+ systems. While the domain is enterprise observability, the core problems are close to quantitative research infrastructure: data readiness, high-volume time-series processing, anomaly detection, baseline comparison, alerting, SLOs, and live monitoring. I also built operational controls around logging, traceability, regression-style checks, runbooks, and escalation paths for critical systems.

My project work is directly aligned with short-term quantitative research. I built a sub-second trading algorithm using live brokerage APIs and intraday market data streams, with momentum and mean-reversion signals, automated risk management, position sizing, and performance monitoring. I also built a Python portfolio risk optimization framework for market data ingestion, backtesting, Sharpe analysis, Monte Carlo simulation, and reproducible quantitative experiments, plus an intraday strategy validation prototype for comparing simulated and live behavior.

Beyond research, I bring strong software ownership. I led the 0-to-1 delivery of a patent-pending platform with reusable components, real-time data rendering, and production workflows, and I led five engineers through code reviews, technical design, testing expectations, documentation, and cross-team standards. I am comfortable presenting results, discussing open questions, and iterating with technical stakeholders under production pressure.

I would welcome the opportunity to bring my Python, statistical analysis, time-series data, backtesting, monitoring, and production systems experience to Squarepoint's CTA/short-term research efforts. Thank you for your consideration, and I would be glad to discuss how my background can support robust automated trading research across liquid markets.

Best regards,



Dhruv Doshi